

UniBank White-Label Banking Platform

Executive Summary

UniBank Engineering

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Platform Overview

UniBank is an enterprise-grade **white-label digital banking platform** purpose-built for the Southern African market. Designed to serve the unbanked and underbanked population, the platform transforms smartphones into full-featured banking instruments, enabling contactless NFC payments, peer-to-peer transfers, bill payments, and agent-assisted cash services — all without requiring physical bank cards or branch visits.

The platform supports **multi-tenant white-label deployments**, allowing financial institutions, mobile network operators, and fintech companies to launch their own branded banking services rapidly, with complete data isolation, configurable fee structures, and independent branding.

Key Value Propositions

- **Zero-cost accounts** for end users — eliminating barriers to financial inclusion
- **NFC-as-Card technology** — smartphones become virtual payment cards at existing POS terminals
- **Agent banking network** — extending financial services through merchant agents earning commissions
- **White-label multi-tenancy** — one platform powering multiple independent banking brands
- **On-premise deployment** — meeting regulatory and data sovereignty requirements

Target Metrics (Year 1)

Metric	Target
Consumer users	500,000
Merchant agents	10,000
Monthly transactions	5,000,000
White-label tenants	3
Payment latency	< 2 seconds
System uptime	99.9%

Architecture Overview

UniBank uses a **Modular Monolith with Satellite Services** architecture, balancing operational simplicity with enterprise-grade security and compliance requirements.

Design Philosophy

The core banking logic runs as a single deployable unit (modular monolith) for transactional consistency and operational simplicity, while security-sensitive and hardware-dependent components run as isolated satellite services. This approach delivers:

- **ACID transactions** across all core banking operations
- **PCI-DSS compliance** through network isolation of payment processing
- **Hardware integration** for HSM cryptography and POS terminal management
- **Manageable complexity** for a focused development team

Technology Stack

Component	Technology	Purpose
Backend	.NET 10 / C#	Core services, high performance
Mobile	Kotlin Multiplatform	Shared logic, native Android/iOS UI
API Protocol	gRPC + Protocol Buffers	Binary serialization, strong contracts
Database	PostgreSQL 18	ACID-compliant, zero licensing cost
Cache	Redis	Hot data, session management
Messaging	Wolverine + MQTT	Async processing, terminal comms
Admin UI	Blazor Server + MudBlazor	Real-time back-office portal
Monitoring	Prometheus + Grafana + ELK	Metrics, dashboards, log aggregation
Deployment	Docker Compose	On-premise, reproducible environments
CI/CD	GitLab CI / Jenkins	Automated build, test, deploy pipeline

System Modules

1. API Gateway

The API Gateway is the single entry point for all client requests, providing a unified security perimeter and traffic management layer.

Capabilities:

- JWT authentication and token validation
- Multi-tenant context extraction and routing
- Per-user and per-tenant rate limiting via Redis
- Request/response logging with PII masking
- gRPC server reflection for client discovery
- Health check and readiness endpoints
- Prometheus metrics exposure

The gateway enforces that every request carries a valid tenant context, ensuring complete data isolation from the network edge.

2. Accounts Module

The Accounts module manages the complete user lifecycle from registration through daily account operations.

Capabilities:

- **Self-registration** via phone number with OTP verification
- **PIN creation** and secure management (bcrypt hashed)
- **Biometric authentication** (fingerprint/face) with PIN fallback
- **Device binding** — sessions tied to specific devices for security
- **Device transfer** — account portability when users change phones
- **Multi-device session management** with concurrent session controls
- **Account status management** (Active, Suspended, Frozen, Closed)
- **Lockout protection** — automatic lockout after 3 failed authentication attempts
- **Balance inquiry** with Redis-cached real-time balances
- **Transaction history** with filtering and pagination

3. KYC (Know Your Customer) Module

The KYC module implements regulatory-compliant identity verification using document and biometric checks.

Capabilities:

- **Document upload** — national ID, passport, or driver's license (streaming upload)
- **Selfie capture** with liveness detection token
- **Facial recognition matching** between selfie and ID document photo
- **Tiered KYC levels** — progressive verification unlocking higher transaction limits
- **Automated verification** with manual review fallback
- **Admin review workflow** — approve, reject, or request resubmission

4. Payments Module

The Payments module is the platform's core differentiator, enabling smartphone-based contactless payments at existing POS infrastructure.

Capabilities:

- **NFC contactless payments** — tap phone on POS terminal to pay (Host Card Emulation)
- **Payment tokenization** — virtual card credentials stored securely, real card data never exposed
- **EMV QR code generation** — merchants generate scannable payment codes
- **QR code payment processing** — customers scan to pay, supporting both static and dynamic QR
- **High-value transaction authorization** — PIN verification for amounts above configurable thresholds
- **Real-time payment notifications** — server-streaming gRPC for instant transaction alerts
- **HSM-backed cryptography** — all card operations secured through Hardware Security Module

5. Transfers Module

The Transfers module enables domestic and international money movement between accounts.

Capabilities:

- **P2P transfers** — send money to any registered user by phone number
- **Cross-border remittances** — international transfers with corridor-based routing
- **Real-time exchange rates** — transparent currency conversion with live rates
- **Transfer status tracking** — full lifecycle visibility from initiation to completion
- **Delivery time estimation** — corridor-specific expected delivery windows
- **Multi-currency support** — ZWG (Zimbabwe Gold) as home currency with regional currency support

6. Bill Payments Module

The Bill Payments module connects users to utility and service providers for bill settlement.

Capabilities:

- **Provider catalog** — electricity, water, telecommunications, and other utility billers
- **Bill payment processing** — pay any registered provider by reference number
- **Saved billers** — store frequently used providers and references for quick repeat payments
- **Provider discovery** — search and filter by category and country
- **Payment history** — complete bill payment records with receipt data

7. Merchants Module

The Merchants module supports onboarding and ongoing management of businesses accepting UniBank payments.

Capabilities:

- **Merchant registration** — business details, location, and type classification
- **Business document upload** — KYC for merchants (business registration, tax certificates)
- **Profile management** — update business information, contact details, operating hours
- **Settlement processing** — automated periodic payouts of collected funds
- **Commission tracking** — per-transaction commission calculation and reporting
- **Transaction history** — full merchant transaction records with export capability

8. Agent Banking Module

The Agent module extends financial services through a network of merchant agents who perform cash services on behalf of the platform.

Capabilities:

- **Cash-in operations** — agents accept physical cash and deposit to customer accounts
- **Cash-out operations** — agents dispense cash against customer account withdrawals
- **Dual PIN verification** — both agent and customer PIN required for cash transactions
- **Float management** — real-time agent working capital tracking with threshold alerts
- **Float limit enforcement** — prevents transactions exceeding available float
- **Commission reporting** — per-transaction commission visibility and payout tracking
- **Transaction receipts** — printable confirmation for both agent and customer

9. Switching Server

The Switching Server is a network-isolated satellite service that integrates with the national payment infrastructure.

Capabilities:

- **ISO 8583 integration** — standard ATM/POS protocol for legacy switch connectivity
- **ISO 20022 integration** — modern XML-based banking standard for interbank transfers
- **Outbound transaction routing** — route payments to acquiring banks and institutions
- **Inbound response handling** — process authorization responses from the national switch
- **Daily reconciliation** — automated matching of sent vs. settled transactions
- **Error handling and retries** — resilient message delivery with configurable retry policies
- **Redundant connectivity** — dual connections to the national switch backbone for high availability

10. Terminal Manager

The Terminal Manager is a satellite service that manages the lifecycle of EFT POS terminals deployed at merchant locations.

Capabilities:

- **Terminal registration** — onboard new POS devices with serial number and model tracking
- **MQTT-based communication** — lightweight, reliable messaging to terminals in the field
- **Firmware distribution** — push firmware updates to terminals over the air
- **Encryption key provisioning** — secure key injection and rotation via HSM
- **Configuration management** — push configuration changes (merchant details, parameters)
- **Status monitoring** — real-time terminal health (online, offline, decommissioned)
- **Heartbeat tracking** — detect and alert on unresponsive terminals

11. HSM Service

The HSM (Hardware Security Module) Service is a network-isolated satellite service providing cryptographic operations for payment security.

Capabilities:

- **Key generation and storage** — create and manage encryption keys within the HSM
- **Session key derivation** — derive per-transaction keys from master keys
- **PIN block encryption/decryption** — ISO 0, ISO 3, and ISO 4 PIN block formats
- **Message authentication codes** — HMAC-SHA256, CMAC-AES, CBC-MAC generation and verification
- **Payment tokenization** — generate tokens for card credentials (NFC payments)
- **PKCS#11 interop** — standard hardware abstraction for physical and virtual HSM devices
- **Key rotation** — scheduled rotation of encryption keys per PCI-DSS requirements

12. Notification Service

The Notification Service delivers real-time alerts across multiple channels.

Capabilities:

- **Push notifications** — Firebase Cloud Messaging (Android) and APNs (iOS)
- **SMS delivery** — integration with SMS gateway for critical alerts
- **In-app notifications** — persistent notification history within the mobile app
- **Template management** — configurable notification templates per event type
- **Delivery tracking** — monitor delivery status with retry on failure
- **PII masking** — sensitive data masked in notification content
- **Event-driven** — triggered automatically by domain events (transactions, auth, KYC, fraud)

13. Reporting Engine

The Reporting Engine aggregates platform data into actionable business intelligence.

Capabilities:

- **Executive dashboard** — real-time metrics (users, transactions, revenue, active merchants)
- **User growth reports** — registration trends, churn analysis, daily/monthly active users
- **Merchant performance** — transaction volumes, settlement history, commission payouts
- **Revenue reporting** — breakdown by transaction type, currency, and tenant
- **Reconciliation reports** — daily matching against the national switch
- **Export functionality** — CSV, PDF, and Excel report generation (streaming download)
- **Configurable granularity** — daily, weekly, monthly aggregation windows

14. Admin Portal

The Admin Portal is the back-office web application for platform operations, support, and configuration.

Capabilities:

- **Customer management** — search, view, suspend, freeze, or close user accounts
- **Merchant management** — approve, suspend, or close merchant accounts
- **KYC review workflow** — review identity documents, approve or reject verification
- **Transaction search** — query transactions by account, date, status, type, or amount
- **Dispute management** — create, track, and resolve chargebacks and disputes
- **Fraud alert review** — review flagged transactions, take action, suspend accounts
- **System configuration** — branding, fees, limits, and commission rates per tenant
- **Reporting dashboards** — visual analytics with drill-down capabilities
- **Audit log** — searchable history of all administrative actions
- **Real-time updates** — SignalR-based live data refresh

15. Fraud Detection Module

The Fraud Detection module provides real-time transaction monitoring and anomaly detection.

Capabilities:

- **Velocity checks** — detect abnormal transaction frequency per account or merchant
- **Geolocation analysis** — identify impossible travel patterns between transactions
- **Device fingerprinting** — alert on transactions from unrecognized devices
- **Amount threshold monitoring** — flag unusually large or atypical transaction amounts
- **Risk scoring** — real-time fraud score calculation for every transaction
- **Alert generation** — automatic fraud alerts escalated to admin review queue
- **Account suspension** — immediate account freeze capability for confirmed fraud

16. White-Label & Multi-Tenancy Module

The White-Label module enables the platform to serve multiple independent banking brands from a single deployment.

Capabilities:

- **Branding configuration** — custom logos, color schemes, CSS, and support contacts per tenant
- **Fee structure management** — configurable fees per transaction type per tenant
- **Transaction limits** — per-transaction, daily, and monthly limits per tenant
- **Commission rates** — customizable agent and merchant commission structures
- **Schema-per-tenant isolation** — each tenant's data in a separate database schema
- **Tenant-scoped configuration** — independent settings that do not affect other tenants

17. Mobile Application

The mobile app is the primary user interface, built with Kotlin Multiplatform for code sharing and native performance.

Capabilities:

- **Cross-platform** — single shared codebase for Android (Jetpack Compose) and iOS (SwiftUI)
- **NFC Host Card Emulation** — tap-to-pay at any contactless POS terminal
- **QR code scanning** — camera-based EMV QR payment scanning
- **Biometric authentication** — fingerprint and face recognition
- **Offline transaction queuing** — queue transactions when connectivity is intermittent
- **Push notification handling** — FCM (Android) and APNs (iOS) integration
- **White-label branding** — dynamic theming based on tenant configuration
- **Secure gRPC communication** — TLS-encrypted binary protocol to the backend

Security & Compliance

Multi-Tenancy & Data Isolation

- **Schema-per-tenant** database isolation — each operator's data is physically separated
- JWT claims enforce tenant context on every API call
- Row-level security policies prevent cross-tenant data access at the database layer
- Tenant secrets (API keys, encryption keys) stored encrypted at rest

Encryption & Key Management

- **TLS 1.3** on all communication channels (gRPC, MQTT, HTTPS)
- **AES-256** encryption for sensitive data at rest (PII, credentials, documents)
- PINs hashed with **bcrypt** — never logged or transmitted in plaintext
- All payment cryptography routed through the **Hardware Security Module**
- Automated key rotation on configurable schedules

Authentication & Authorization

- **JWT tokens** — 15-minute access tokens, 7-day refresh tokens
- **Device binding** — sessions tied to device identifiers, preventing token theft
- **Lockout protection** — 30-minute lockout after 3 failed attempts
- **Role-Based Access Control** — admin portal roles: viewer, support, admin, super-admin
- **Comprehensive audit logging** — all actions logged with actor, timestamp, and result

PCI-DSS Compliance

- Payment card data is **never stored** — tokenization only
- Switching and HSM services deployed on **isolated network segments**
- Restricted access to payment processing systems
- All card-related cryptographic operations performed within the HSM
- Complete transaction audit trail maintained

Development Summary

Project Scope

The platform was delivered across **8 development sprints**, comprising **76 user stories** totaling **362 story points**.

Sprint	Focus Area	Stories	Points
1	Foundation & scaffolding	11	62
2	KYC, authentication, device binding	12	49
3	NFC payments, QR codes, tokenization	8	45
4	P2P transfers, agent cash services, bill pay	11	47
5	Payment switch integration, reconciliation	9	47
6	Terminal management, white-label configuration	7	29
7	Admin portal, reporting, dispute management	13	58
8	Fraud detection, security hardening	5	25
Total		76	362

Testing Strategy

- **Unit tests** — domain logic, command/query handlers, validation rules
- **Integration tests** — end-to-end API flows with TestContainers (PostgreSQL, Redis, MQTT)
- **Automated CI/CD** — build, test, and container image generation on every commit

Module Summary

Module	Type	Purpose
API Gateway	Service	Authentication, routing, rate limiting
Accounts	Core	Registration, auth, profiles, sessions
KYC	Core	Identity verification, document review
Payments	Core	NFC, QR code, tokenization
Transfers	Core	P2P, cross-border, exchange rates
Bill Payments	Core	Utility and service bill settlement
Merchants	Core	Merchant onboarding, settlement
Agent Banking	Core	Cash-in/out, float management
Fraud Detection	Core	Anomaly detection, risk scoring
White-Label	Core	Branding, fees, limits, multi-tenancy
Admin	Core	Back-office operations, disputes
Switching Server	Satellite	ISO 8583/20022 switch integration
Terminal Manager	Satellite	MQTT-based POS lifecycle
HSM Service	Satellite	PKCS#11 cryptographic operations
Notifications	Satellite	Push, SMS, in-app alerts
Reporting Engine	Satellite	Analytics, dashboards, exports
Admin Portal	Application	Blazor web back-office
Mobile App	Client	KMP Android/iOS application

Conclusion

UniBank delivers a **complete digital banking ecosystem** addressing the specific needs of the Southern African market:

1. **Financial Inclusion** — zero-cost accounts and agent networks reach the unbanked population where traditional banking infrastructure does not exist.
2. **NFC Innovation** — transforming smartphones into virtual payment cards eliminates the need for costly card issuance while leveraging existing POS terminal infrastructure.
3. **White-Label Scalability** — a single platform deployment supports multiple independent banking brands with complete data isolation, enabling rapid market expansion.
4. **Agent Ecosystem** — a self-sustaining network of merchant agents earns commissions for providing cash services, organically expanding the platform's physical reach.
5. **Regulatory Compliance** — schema-per-tenant data isolation, HSM-based cryptography, PCI-DSS network segmentation, and comprehensive audit logging meet the requirements of banking regulators.
6. **On-Premise Sovereignty** — Docker-based on-premise deployment ensures data residency compliance and operational control for regulated financial environments.

The platform is architecturally sound, security-hardened, and ready for pilot deployment targeting **500,000 users** and **10,000 merchant agents** in its first year of operation.

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